

Database Lab Session 1: Database Modeling and Relational Normalization

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The goal of this practical session is to diagnose, analyze, and normalize a realistic dataset provided directly from public health administration exports. You will evaluate structural integrity, extract functional dependencies based on domain rules, identify potential runtime anomalies, and apply database decomposition algorithms to transition the schema into a robust Third Normal Form (3NF).

The Universal Schema under Review Consider the flat universal relation tracking adverse pharmaceutical side effects:

FDA_Adverse_Events(
report_id, receive_date, year, month, quarter, serious, serious_flags, is_fatal, is_hospitalized,
is_life_threat, is_disabling, reactions, primary_reaction, reaction_outcomes, patient_recovered,
num_reactions, suspect_drug, brand_name, drug_route, drug_indication, manufacturer,
pharm_class, num_drugs, drug_count_category, patient_age_years, age_group, patient_sex,
patient_weight_kg, country, report_age_days)

0.1 Dataset Exploration and First Normal Form

Before writing dependencies, an engineer must address the basic physical requirements of relational modeling.

1. **Question 1.1:** In a standard medical tracking scenario, an individual patient often experiences multiple side effects simultaneously (e.g., a patient suffers from "*Nausea, Dizziness, Acute Migraine*"). If the column **reactions** stores these items as a single comma-separated string or array value within a single row, does this table comply with **First Normal Form (1NF)**? Justify your answer using the formal definition of 1NF.
2. **Question 1.2:** State the operational and querying penalties encountered by software development teams when a database relies on non-atomic text fields like multi-valued strings.
3. **Assumption for subsequent phases:** For the remainder of this lab, assume that the values have been flattened into atomic items. To preserve the uniqueness of rows under a single event where a patient has multiple reactions, the identifier **report_id** must be paired as a compound key: [report_id, primary_reaction].

0.2 Functional Dependency (FD) Extraction

Functional dependencies cannot be derived through statistical evaluation of row snapshots; they reflect static, immutable realities of the underlying domain logic.

1. **Question 2.1:** Using structural constraints of chronological time, look at the group of attributes: `{receive_date, year, month, quarter}`. State and mathematically write out the functional dependency governing this subgroup. Justify its invariant nature.
2. **Question 2.2:** Evaluate the drug catalog attributes: `{suspect_drug, brand_name, manufacturer, pharm_class}`. Write the non-trivial functional dependencies stemming from the fact that a drug registration code is unique across commercial distribution.
3. **Question 2.3:** The medical agency segments clinical analytics via age cohorts. Propose a functional dependency matching the interaction between `patient_age_years` and `age_group`.
4. **Question 2.4:** Compile your findings and formally write out the set F of all dominant structural non-trivial Functional Dependencies for the entire universal relation.

0.3 Normal Form Diagnosis & Systemic Anomalies

Using your extracted dependencies F and using the compound base primary key, you will evaluate systemic structural vulnerability.

1. **Question 3.1:** Is the universal relation in **Second Normal Form (2NF)**? Point out specific functional dependencies that validate or invalidate compliance.
2. **Question 3.2:** Is the schema compliant with **Third Normal Form (3NF)**? (Identify any transitive pathways $(\alpha \rightarrow \beta \rightarrow \gamma)$ where non-key paths dictate the data landscape).
3. **Question 3.3:** Describe a concrete real-world scenario illustrating how this unnormalized structure triggers:
 - An **Update Anomaly** when a manufacturing plant is acquired or changes names.
 - An **Insertion Anomaly** when a new drug formula receives immediate health approval but has not yet caused a medical emergency.
 - A **Deletion Anomaly** when historical case logs are archived.

0.4 Lossless 3NF Decomposition

Using Heath's Theorem, an unadapted relation can be safely split across valid functional dependency boundaries without generating spurious data rows.

1. **Question 4.1:** Apply Heath's Theorem progressively to isolate independent static domains. Provide the complete schema layout for the final decoupled normalized database. Considering that the only FDs are those defined in the previous questions.
2. **Question 4.2:** For each resulting relation schema, explicitly mark and verify:
 - The **Primary Key (PK)**.
 - The **Foreign Keys (FK)** establishing relational referential integrity back to the target schemas.